

ESM 262



Assignment #5

Modularity

With a partner

Think of an example complex data science task/program that you might want to design with:

- at least 3 modules
- 1 module must be used at least twice
- at least one *for* (or *while*) loop

Planning - Step 1

- sketch a diagram that shows how these modules would be put together to go from inputs to output for the overall program (30pts)
- write a contract for the overall program (input, output, parameters) (10pts)
- break the program in to modules (naming them appropriately), write a contract for each one (20pts)

Important - Check in with myself or Ojas to get the OK on your plan BEFORE you go to Step 2

Implementation - Step 2

- write functions (stored in separate documents) for each module (30pts)
- write an overall program that manages *calls* to each function (10pts)
- create some data and use it to test the program (10pts)

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